



Spring/Fall 2027 Ph.D. Positions in Transportation Engineering at Rutgers University

Dr. Keke Long is recruiting 2-3 fully funded Ph.D. students for Spring or Fall 2027 admission at Rutgers University. Students with backgrounds in transportation, mathematics, computer science, mechanical engineering or related fields are welcome to apply. Please email your CV and a brief description of your relevant experience to klongcee@gmail.com, using the subject line 'PhD Application_Spring/Fall 2027_Name'. Master's students and visiting students are also welcome to reach out.

Program Introduction: <https://grad.rutgers.edu/academics/programs/civil-environmental-engineering>

Application Requirements: <https://cee.rutgers.edu/graduate-applicant-criteria>

Research Interests

1. **Physics-Enhanced AI for Transportation:** Develop interpretable, physically consistent, and robust AI methods for transportation systems.
2. **Evaluation of CAVs:** Evaluate and improve the safety and performance of CAVs in challenging scenarios, including extreme weather, long-tail events, and interactions with vulnerable road users.
3. **Intelligent Transportation Systems:** Study how people, vehicles, and transportation infrastructure can be connected and coordinated to improve safety and efficiency.

About the Advisor

Dr. Long will join Rutgers University as an Assistant Professor in Civil and Environmental Engineering. She is currently a Research Associate at the University of Wisconsin–Madison. She has served as co-PI on federally funded transportation research project and instructor of CCAT certificate program. Dr. Long studies how to improve the safety and reliability of transportation systems. Her research develops AI-based methods that use physical knowledge to improve interpretability and adaptation to changing conditions. She validates these methods through simulation, digital twins, and real-vehicle experiments to support the reliable deployment of automated transportation systems. She has published over 30 papers including Transportation Research Part A/C/E and IEEE ITS/ITSM.

About Rutgers University

Rutgers–New Brunswick is a highly ranked public university (e.g., 42nd nationally and 16th among public universities @ U.S. News) with a reputable engineering program (e.g., 27th among public university engineering programs, with Civil and Environmental Engineering ranked in the top 50 nationally). It is also home to the Center for Advanced Infrastructure and Transportation (CAIT), a U.S. DOT University Transportation Center.

Rutgers–New Brunswick locates at the center part of New Jersey, the “garden state”, with convenient access to New York City and Philadelphia. The surrounding region offers students access to technology companies, internship opportunities, and a diverse community.